

Git Fundamentals Assignment

Setup

1. Install Git if you have not already, and confirm your version with a command of your choice.
2. Configure your identity (name and email) so your commits are attributed to you.
3. Create a new folder called git-basics-lab and turn it into a Git repository.

Reflection: What actually happens on your filesystem when you run the command that turns a folder into a Git repository? Look inside the hidden folder Git creates (hint: on most systems, folders starting with a dot are hidden, use `ls -a` or enable "show hidden files"). What do you find there?

```
(kali㉿kali)-[~]
└─$ git --version
git version 2.53.0

(kali㉿kali)-[~]
└─$ git config user.name user.email

(kali㉿kali)-[~]
└─$ git config user.name Hatman025

(kali㉿kali)-[~]
└─$ git config --global user.name Hatman025

(kali㉿kali)-[~]
└─$ git config --global user.email tinawo2500@gmail.com

(kali㉿kali)-[~]
└─$ mkdir git-basics-lab

(kali㉿kali)-[~]
└─$ cd git-basics-lab

(kali㉿kali)-[~/git-basics-lab]
└─$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/kali/git-basics-lab/.git/

(kali㉿kali)-[~/git-basics-lab]
└─$ ls -a
.  ..  .git
```

Part 1

Git famously works with three areas: the working directory, the staging area (index), and the repository (commit history).

4. Create a file called notes.txt with a single line of text. Check the status of your repository. What does Git say about this file?
5. Stage the file. Check status again. What changed in the wording of the output?
6. Commit the file with a meaningful message.
7. Modify notes.txt by adding a second line. Check status again, before staging.

Reflection: Explain, in your own words, the difference between a file that is "untracked," "modified," and "staged." Why does Git separate staging from committing at all? What workflow problem does this solve?

8. Try committing your change from step 7 without staging it first. What happens? Now stage and commit properly.
 - An untracked file means that the file wasn't created with the 'git add' command and can't be prepared to be committed
 - A modified file means that some changes has been done in the file, therefore has to be added using the 'git add' prompt again
 - A staged file means that the file is prepared to be committed

```
kali@kali: ~/git-basics-lab
Session Actions Edit View Help

(kali@kali)-[~/git-basics-lab/.git]
$ cat <<'EOF' > note.txt
heredoc> This is my single line of text
heredoc> EOF
(kali@kali)-[~/git-basics-lab/.git]
$ rm -rf note.txt
(kali@kali)-[~/git-basics-lab/.git]
$ cd .
(kali@kali)-[~/git-basics-lab/.git]
$ cd ..
(kali@kali)-[~/git-basics-lab]
$ cat <<'EOF' > note.txt
heredoc> This is my single line of text
heredoc> EOF
(kali@kali)-[~/git-basics-lab]
$ ls al
ls: cannot access 'al': No such file or directory
(kali@kali)-[~/git-basics-lab]
$ ls -la
total 16
drwxrwxr-x  3 kali kali 4096 Jul 11 19:31 .
drwx----- 43 kali kali 4096 Jul 11 19:21 ..
drwxrwxr-x  6 kali kali 4096 Jul 11 19:30 .git
-rw-rw-r--  1 kali kali   31 Jul 11 19:31 note.txt
(kali@kali)-[~/git-basics-lab]
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    note.txt

nothing added to commit but untracked files present (use "git add" to track)
(kali@kali)-[~/git-basics-lab]
$
```

```
kali@kali: ~/git-basics-lab

Session Actions Edit View Help

(kali@kali)-[~/git-basics-lab]
$ git add note.txt

(kali@kali)-[~/git-basics-lab]
$ git status
On branch master
No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   note.txt

you solved the lab!
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LAB Solved
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(kali@kali)-[~/git-basics-lab]
$ git commit -m "Add note.txt with specific project content"
[master (root-commit) 4bdf35d] Add note.txt with specific project content
1 file changed, 1 insertion(+)
create mode 100644 note.txt

(kali@kali)-[~/git-basics-lab]
$ echo "This is the actual beginning" >> note.txt

(kali@kali)-[~/git-basics-lab]
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   note.txt

no changes added to commit (use "git add" and/or "git commit -a")

(kali@kali)-[~/git-basics-lab]
$ git commit -m "Add note.txt with specific project content"
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   note.txt

no changes added to commit (use "git add" and/or "git commit -a")
```

```
(kali@kali)-[~/git-basics-lab]
$ git commit -m "Add note.txt with specific project content"
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   note.txt

no changes added to commit (use "git add" and/or "git commit -a")

(kali@kali)-[~/git-basics-lab]
$ git add note.txt

(kali@kali)-[~/git-basics-lab]
$ git commit -m "Add note.txt with specific project content"
[master 6d4ee6b] Add note.txt with specific project content
1 file changed, 1 insertion(+)

(kali@kali)-[~/git-basics-lab]
$
```

Part 2: Exploring History

9. Make three more small commits to notes.txt, each with a distinct, clear commit message.
10. View your commit history in at least two different formats or levels of detail.
11. Find the unique identifier (hash) of your second commit.
12. Use that hash to view exactly what changed in that specific commit.

Reflection: Why does Git use hashes instead of sequential numbers (like commit 1, commit 2, commit 3) to identify commits? What does this tell you about how Git is designed to work in a distributed, multi-person environment?

13. Compare the difference between your working directory right now and your very first commit.
 - Git uses hashes instead of sequential numbers because each commit is identified by a hash of its contents and metadata. This means the commit ID is tied to the exact state of the repository at that point. If any changes are made to the file, the hash changes. Sequential numbers wouldn't provide this guarantee.
 - Git is designed to work in a distributed, multi-person environment because of its decentralization, collaboration and tamper resistance features. Git is designed for a distributed environment where many people work independently. If everyone used sequential numbers, conflicts would be inevitable. Hashes avoid this problem because they're globally unique.

```
kali@kali: ~/git-basics-lab
Session Actions Edit View Help
└─$ echo "This writeup is about real world incident" >> note.txt

(kali@kali)-[~/git-basics-lab]
└─$ git add note.txt
mode execution via web shell upload LAB Solved

(kali@kali)-[~/git-basics-lab]
└─$ git commit -m "Add note.txt with the facts of the writeup"
[master 9ca9db2] Add note.txt with the facts of the writeup
1 file changed, 1 insertion(+)

(kali@kali)-[~/git-basics-lab]
└─$ echo "The world is full of whao moments" >> note.txt
you solved this lab! Share your skills! Twitter LinkedIn Continue

(kali@kali)-[~/git-basics-lab]
└─$ git add note.txt

(kali@kali)-[~/git-basics-lab]
└─$ git commit -m "Add note.txt with the description of the content"
[master 04c1ad0] Add note.txt with the description of the content
1 file changed, 1 insertion(+)

(kali@kali)-[~/git-basics-lab]
└─$ echo "The moments are full of surprises" >> note.txt

(kali@kali)-[~/git-basics-lab]
└─$ git add note.txt

(kali@kali)-[~/git-basics-lab]
└─$ git commit -m "Add note.txt with stating the actual purpose of the writeup"
[master 6c4b704] Add note.txt with stating the actual purpose of the writeup
1 file changed, 1 insertion(+)

(kali@kali)-[~/git-basics-lab]
└─$ git log
commit 6c4b70474abf8ce08ca649b3f828deb3ce8 (HEAD -> master)
Author: Hatman025 <tinawo2500@gmail.com>
Date: Sat Jul 11 20:11:56 2026 -0400

    Add note.txt with stating the actual purpose of the writeup

commit 04c1ad0a205e51e8b3c1d5bd73fdf02c85950829
Author: Hatman025 <tinawo2500@gmail.com>
Date: Sat Jul 11 20:10:51 2026 -0400

    Add note.txt with the description of the content

Home | My account
```

```
kali@kali: ~/git-basics-lab
Session Actions Edit View Help
(kali@kali)-[~/git-basics-lab]
$ echo "The moments are full of surprises" >> note.txt
(kali@kali)-[~/git-basics-lab]
$ git add note.txt
(kali@kali)-[~/git-basics-lab]
$ git commit -m "Add note.txt with stating the actual purpose of the writeup"
[master 6c4b704] Add note.txt with stating the actual purpose of the writeup
1 file changed, 1 insertion(+)
(kali@kali)-[~/git-basics-lab]
$ git log
commit 6c4b70474abf8e98f8ce08ca649b3f828deb3ce8 (HEAD -> master)
Author: Hatman025 <tinawo2500@gmail.com>
Date: Sat Jul 11 20:11:56 2026 -0400
    Add note.txt with stating the actual purpose of the writeup
commit 04c1ad0a205e51e8b3c1d5bd73fdf02c85950829
Author: Hatman025 <tinawo2500@gmail.com>
Date: Sat Jul 11 20:10:51 2026 -0400
    Add note.txt with the description of the content
commit 9ca9db2439877045424d133b4199d6f23310cdea
Author: Hatman025 <tinawo2500@gmail.com>
Date: Sat Jul 11 20:09:57 2026 -0400
    Add note.txt with the facts of the writeup
commit 6d4ee6bc970bc42106c0d97939842231e9bf6d17
Author: Hatman025 <tinawo2500@gmail.com>
Date: Sat Jul 11 19:57:23 2026 -0400
    Add note.txt with specific project content
commit 4bdf35daf4fd15bc87f4f89a10345aed75688449
Author: Hatman025 <tinawo2500@gmail.com>
Date: Sat Jul 11 19:51:23 2026 -0400
    Add note.txt with specific project content
(kali@kali)-[~/git-basics-lab]
$
```



```
kali@kali: ~/git-basics-lab
Project
Session Actions Edit View Help

--(kali@kali)-[~/git-basics-lab]
$ git log --online
fatal: unrecognized argument: --online via web shell upload LAB Solved

--(kali@kali)-[~/git-basics-lab]
$ git log --oneline
c4b704 (HEAD -> master) Add note.txt with stating the actual purpose of the writeup
4c1ad0 Add note.txt with the description of the content
ca9db2 Add note.txt with the facts of the writeup
d4ee6b Add note.txt with specific project content
bdf35d Add note.txt with specific project content

--(kali@kali)-[~/git-basics-lab]
$ git log --stat
commit 6c4b70474abf8ce08ca649b3f828deb3ce8 (HEAD -> master)
author: Hatman025 <tinawo2500@gmail.com>
date: Sat Jul 11 20:11:56 2026 -0400

    Add note.txt with stating the actual purpose of the writeup

note.txt | 1 +
1 file changed, 1 insertion(+)

commit 04c1ad0a205e51e8b3c1d5bd73fd02c85950829
author: Hatman025 <tinawo2500@gmail.com>
date: Sat Jul 11 20:10:51 2026 -0400

    Add note.txt with the description of the content

note.txt | 1 +
1 file changed, 1 insertion(+)

commit 9ca9db2439877045424d133b4199d6f23310cdea
author: Hatman025 <tinawo2500@gmail.com>
date: Sat Jul 11 20:09:57 2026 -0400

    Add note.txt with the facts of the writeup

note.txt | 1 +
1 file changed, 1 insertion(+)

commit 6d4ee6bc970bc42106c0d97939842231e9bf6d17
author: Hatman025 <tinawo2500@gmail.com>
date: Sat Jul 11 19:57:23 2026 -0400
```



```
(kali@kali)-[~/git-basics-lab]
$ git log --oneline --graph --all
* 6c4b704 (HEAD -> master) Add note.txt with stating the actual purpose of the writeup
* 04c1ad0 Add note.txt with the description of the content
* 9ca9db2 Add note.txt with the facts of the writeup
* 6d4ee6b Add note.txt with specific project content
* 4bdf35d Add note.txt with specific project content

(kali@kali)-[~/git-basics-lab]
$ git log --graph
* commit 6c4b70474abfbe98f8ce08ca649b3f828deb3ce8 (HEAD -> master)
  Author: Hatman025 <tinawo2500@gmail.com>
  Date: Sat Jul 11 20:11:56 2026 -0400

    Add note.txt with stating the actual purpose of the writeup

* commit 04c1ad0a205e51e8b3c1d5bd73fdf02c85950829
  Author: Hatman025 <tinawo2500@gmail.com>
  Date: Sat Jul 11 20:10:51 2026 -0400

    Add note.txt with the description of the content

* commit 9ca9db2439877045424d133b4199d6f23310cdea
  Author: Hatman025 <tinawo2500@gmail.com>
  Date: Sat Jul 11 20:09:57 2026 -0400

    Add note.txt with the facts of the writeup

* commit 6d4ee6bc970bc42106c0d97939842231e9bf6d17
  Author: Hatman025 <tinawo2500@gmail.com>
  Date: Sat Jul 11 19:57:23 2026 -0400

    Add note.txt with specific project content

* commit 4bdf35daf4fd15bc87f4f89a10345aed75688449
  Author: Hatman025 <tinawo2500@gmail.com>
  Date: Sat Jul 11 19:51:23 2026 -0400

    Add note.txt with specific project content

(kali@kali)-[~/git-basics-lab]
$
```

```
(kali@kali)-[~/git-basics-lab]
$ git show 04c1ad0a205e51e8b3c1d5bd73fdf02c85950829
commit 04c1ad0a205e51e8b3c1d5bd73fdf02c85950829
Author: Hatman025 <tinawo2500@gmail.com>
Date: Sat Jul 11 20:10:51 2026 -0400

    Add note.txt with the description of the content

diff --git a/note.txt b/note.txt
index 8e55608..22bff88 100644
--- a/note.txt
+++ b/note.txt
@@ -1,3 +1,4 @@
  This is my single line of text
  This is the actual beginning
  This writeup is about real world incident
+The world is full of whao moments

(kali@kali)-[~/git-basics-lab]
$
```

Session Actions Edit View Help

```
(kali@kali)~/git-basics-lab  
$ git rev-list --max-parents=0 HEAD
```

4bdf35daf4fd15bc87f4f89a10345aed75688449

```
(kali@kali)~/git-basics-lab  
$ git diff 4bdf35daf4fd15bc87f4f89a10345aed75688449
```

```
diff --git a/note.txt b/note.txt  
index efcdb8f..72be847 100644  
--- a/note.txt  
+++ b/note.txt  
@@ -1,5 @@  
  This is my single line of text  
+This is the actual beginning  
+This writeup is about real world incident  
+The world is full of whao moments  
+The moments are full of surprises
```

```
(kali@kali)~/git-basics-lab  
$
```